



Sarthak Consul
Electrical Engineering
Indian Institute of Technology Bombay

16D100012
UG Third Year (B.Tech)
Male
DOB: 30 January 1998

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2019	9.65
Intermediate/+2	CBSE	Jayshree Periwal High School	2016	92.80
Matriculation	CBSE	Step by Step High School	2014	10.00

Pursuing an **Honours** degree in **Electrical Engineering** and a **Minor** in **Computer Science and Engineering**

Academic Achievements

- o Awarded **Advanced Performer** grade (AP) in **4 courses**: *Biology* (15 out of 445), *Introduction to Electrical Systems* (2 out of 149), *Network Theory* (10 out of 143), and *Electronic Devices Lab* (3 out of 139)
- o Secured All India Rank 840 (out of around 200,000 candidates) in JEE Advanced [2016]
- o Recipient of KVPY Fellowship (**AIR 16**) by Dept. of Science and Technology, Govt. of India [2014]
- o Placed in **National Top 1%** in National Standard Examination in Junior Sciences [2012]
- o Ranked **4th** in State Science Talent Search Examination (SSTSE), conducted by Govt. of Rajasthan [2014]
- o Received **Certificate of Merit** for obtaining Grade 'A1' for all 5 subjects in Class X [2014]
- o Recipient of **NTSE Scholarship** by National Council of Research and Training [2012]

Research Experience

Screening of Precancerous Lesions in the Oral Cavity

Medical Deep learning and AI Lab

Guide: Prof. Amit Sethi, EE, IIT Bombay

July 2018 - Present

- o Working on **real-time automatic** detection of the oral cavity as a pre-processing step for oral cancer detection
- o Annotated and organised data from around **800 patients** affected by oral cancer
- o Experimenting with various image processing techniques to extract oral cavity area from images

Semantic Segmentation of Medical Images

Medical Deep learning and AI Lab

Guide: Prof. Amit Sethi, EE, IIT Bombay

May 2018 - July 2018

- o Tested ability of Fully Connected Convolutional Networks in segmentation tasks in medical images.
- o Implemented Global Convolution on a fully-connected ResNet backbone using Pytorch for the **segmentation of lungs** from chest x-rays and **nuclei segmentation** from h/he stained histopathological sections of organs
- o Part of a five member team testing various architectures and conventional image processing techniques as post-processing to improve the state of the art results of **nuclei segmentation** to a mean F1 score of **85%**
- o Reviewed research papers on **ReNet and ReSeg, MobileNet, and Siamese Neural Networks**
- o **Mentored** two freshmen in the team on the basics of machine learning and deep learning
- o Work was used to host the MoNuSeg challenge at MICCAI 2018.

Course Projects

Post-Capture Image Refocusing

CS663: *Digital Image Processing*

Prof. Suyash Awate, CSE, IIT Bombay

Autumn 2018 (Ongoing)

- o Working on depth estimation using an **hybrid Minimum Spanning Tree** approach for cost aggregation in stereo matching, benchmarking it with other techniques such as bilateral filtering on the **Middlebury dataset**
- o Applying the generated depth maps to create depth-induced photo effects such as **interactive depth of field rendering** and replication of the **stage lighting effect** of iPhone X's portrait mode

Analog Sound Locator

EE230: *Analog Systems Lab*

Prof. Siddharth Tallur, EE, IIT Bombay

Spring 2018

- o Designed and implemented an analog circuit to locate distant sound sources. The setup mimicked bi-aural mammalian ears, using the phase difference of the received sound signal to locate the source.
- o **Amplitude invariance** was achieved using JFET delay circuits with dynamic phase locking
- o Circuit was compact and had a fast convergence (spatial angle was obtained **within a millisecond**)
- o One of the **winning projects** in the final evaluation (*amongst 68 teams*)

Reaction Game

EE224: *Digital Systems Lab*

Prof. Madhav P. Desai, EE, IIT Bombay

Spring 2018

- o Built a game to measure reaction times to visual stimulus using a Krypton (Altera MAX V) CPLD board
- o Modelled the game as a **finite-state machine** and realised it with **register-transfer level** abstraction in VHDL

- o Incorporated false-react detection, timeouts and debouncing features. Used the **Double Dabble Algorithm** to convert the binary reaction times to decimal. Integrated a LCD controller to give real-time scores on a LCD

Building a Cryptosuite

Prof. Bernard L. Menezes, CSE, IIT Bombay

CS 101: Computer Programming

Autumn 2016

- o Designed a program that encrypted and decrypted alphanumeric messages using **RSA Algorithm**
- o Computed Discrete logarithms using **Baby Step Giant Step Algorithm**, binary and prime field operations, and modulo arithmetic (*addition, multiplication, exponentiation by binary exponentiation, and inverse*)

Teaching Experience

Teaching Assistant

Prof. Swapneel Mahajan, Mathematics, IIT Bombay

MA207: Partial Differential Equations

August 2018 - Present

- o Teaching and resolving doubts for a class of approximately 60 students. Involved in the formulation of examinations, grading of answer scripts and other logistics.

Instructor of Machine Learning Bootcamp

Summer 2018

- o Taught the fundamentals of Machine Learning to a class of approximately 130 students
- o Covered the basics of ML before delving into the theory and implementation of popular algorithms of supervised and unsupervised learning such as **Random Forest Algorithm**, **CNNs**, and **k-means Clustering**.
- o Prepared Jupyter notebooks (in Python), meant to serve as tutorials, for the various algorithms discussed

Technical Skills

Programming Languages - C, C++, MATLAB, Python (with PyTorch, Tensorflow, Keras & OpenCV), VHDL, Assembly and Embedded C, Bash

Web Development - HTML5, CSS3, PHP

Simulation Tools - NgSpice, Simulink, LabVIEW, Wireshark, ns-3

Design Software - AutoCAD, SolidWorks, Photoshop, L^AT_EX

Key Courses Taken

Electrical Engineering

Microprocessors (with Lab)*, Communication Systems (with Lab)*, EM Waves*, **Probability and Random Processes***, Electronic Devices and Circuits (with Lab), **Network Theory**, Microelectronics, **Signals and Systems**, Electrical Machines and Power Electronics (with Lab), **Analog Circuits (with Lab)**, Digital Systems (with Lab), **Digital Signal Processing****, Digital Communications**, **Control Systems****, Power Systems (with Lab)**

Mathematics Courses

Matrix Computations, Linear Algebra, **Data Analysis**, Calculus, Differential Equations, Complex Analysis

Computer Science and Engineering

Machine Learning*, **Digital Image Processing***, Computer Networks, Data Structures and Algorithms

MOOCs

- o **Machine Learning** (Stanford University - CS229 and Coursera)
- o **Convolutional Neural Network for Computer Vision** (Stanford University - CS231n)
- o **Advanced MATLAB for Scientific Computing** (Stanford University - Lagunita) [with Distinction]
- o **Neural Networks for Machine Learning** (University of Toronto - Coursera)
- o **Computational Neuroscience** (University of Washington - Coursera) *

* Indicates courses to be completed by Dec 2018

** Indicates courses to be completed by May 2019

Extra Curricular Activities

- o Completed over **80 hours** of social service in **National Service Scheme**, IIT Bombay
 - Created a biodiversity map and a database of the various species of flora located on campus
 - Involved in converting a Construction Debris Landfill into a nursery and butterfly garden
- o Team captain of Inter-school Competitive Scrabble Team (Reached **Finals** in Plate and Cup Categories) for 3 years
- o Awarded the Certificate for **Distinction** in the Innovation Programme, Singapore in 2009
- o Member of Prefectorial Committee for 3 years in school

Other Achievements

- o Accepted into the **Gifted Education Programme (GEP)**, Singapore (1 % out of 50,000 students)
- o Awarded **Gold** in the National Mathematical Olympiad of Singapore

[2009]